Decision Analysis Theory

Decision Making Under Risk and Uncertainty 1

Decision Problems – New Product Information

- A manufacturer introducing a new product into the marketplace.
  - What will be the reaction of potential customers?
  - How much of the product should be produced?
  - Should the product be test marketed in a small region before deciding upon full distribution?
  - How much advertising is needed to launch the product successfully?

Decision Problems – Financial Investments

- A financial firm investing in securities.
  - Which are the market sectors and individual securities with the best prospects?
  - Where is the economy headed?
  - How about the interest rates?
  - How should these factors affect the investment decisions?

Decision Problems – Government Contract Bidding

- A government contractor bidding on a new contract.
  - What will be the actual costs of the project?
  - Which other companies might be bidding?
  - What are their likely bids?

Decision Problems – Agricultural Planning

- An agricultural firm selecting a mix of crops and livestock for the upcoming season.
  - What will be the weather conditions?
  - Where are prices headed?
  - What will costs be?

Decision Problems – Oil Drilling

- An oil company deciding whether to drill for oil in a particular location.
  - How likely is oil there?
  - How much of it?
  - How deep will they need to drill?
  - Should geologists investigate the site further before drilling?

Decision Making Under Certainty

- The consequence of every alternative or decision choice is known with certainty.
- Alternative that maximizes the well-being or results in the best outcome is selected.
- Example
  - You have $1,000 to invest for a 1-year period.
  - First alternative is to open a savings account paying 5% interest.
  - Second alternative is to invest in a Government Treasury bond paying 8% interest.
  - If both investments are secure and guaranteed, there is a certainty that the bond will pay higher return.
  - Outcome: $80 return in interest.

Decision Making Under Risk

- There are several possible outcomes for each alternative, and the decision maker knows the probability of occurrence of each outcome.
- Decision maker attempts to maximize the well-being .
  - Maximize expected monetary value.
  - Minimize expected loss.
- Example
  - Probability of rolling any chosen number on a dice is 1/6 and is known.

Decision Making Under Uncertainty

- There are several possible outcomes for each alternative, and the decision maker does not know the probability of various outcomes.
- Sometimes it is impossible to assess the probability of success of a new undertaking.
- Example
  - The probability that a new V-type of aircraft will be in service 10 years from now is not known.

Decision Analysis Theory

- Analytic and systematic approach to tackle problems, or in another words, to the study of decision making.

- Although, occasionally, good decisions yield bad results, in the long run, using decision theory will result in successful outcomes.

Decision Analysis Theory – Six Steps

- Six Steps of Decision Making.
  - 1. Clearly define the problem at hand.
  - 2. List the possible alternatives.
  - 3. Identify the possible outcomes or states of nature.
  - 4. List the payoff or profit of each combination of alternatives and outcomes.
  - 5. Select one of the mathematical decision theory models.
  - 6. Apply the model and make your decision.

Decision Analysis Terminology

- Alternative (Action)
  - Course of action, A_i, to be selected by the decision maker from a set of possible courses of action.
  - $A = \{A_1, A_2, \ldots, A_m\}$
- State of Nature (State of World, Future)
  - Situation found at the time the decision alternative is selected.
  - Not under control of the decision making.
  - $S = \{S_1, S_2, \ldots, S_n\}$

Decision Analysis Terminology

- Probability that the State of Nature will occur
  - State of nature, S_j, is observed with probability P_j.
  - If $P_j(S_j)$ are specified, the decision is said to be under risk.
  - If $P_j(S_j)$ are not specified, the decision is said to be under uncertainty.
- Payoff (Reward, Evaluation Measure)
  - Quantitative measure, R_{ij}, of the value to the decision maker of the consequences of the outcome.
  - Can be positive or negative.
- Payoff Table (Reward Table, Decision Evaluation Matrix)
  - Payoffs for each combination of an action and a state of nature.

Decision Analysis Example

- The owner of a manufacturing company needs to decide whether to expand his operations.

Alternative	State of environment	
	Favorable market ($)	Unfavorable market ($)
Construct a large plant	200,000	-180,000
Construct a small plant	100,000	-20,000
Do nothing	0	0

It is important to include all alternatives, including the "do nothing" one.

General Decision Analysis Model

- Formal way of exhibiting the interaction of a finite set of alternatives and a finite set of futures.
  - Depicts the positive and negative results that may occur for each alternative under each possible future.

Alternative A_i	State of Nature S_j / Probability of occurrence $P_j(S_j)$			
	S_1 $P_1(S_1)$	S_2 $P_2(S_2)$	…	S_n $P_n(S_n)$
A_1	R_{11}	R_{12}	…	R_{1n}
A_2	R_{21}	R_{22}	…	R_{2n}
…	…	…	…	…
A_m	R_{m1}	R_{m2}	…	R_{mn}

General Decision Analysis Model

- Evaluation measures can be either quantitative or qualitative.
- Possible futures not identified can significantly affect the actual outcome relative to the planned outcome.
- Occurrence of one future precludes the occurrence of any other future (mutually exclusive).
- Occurrence of a specific future is not influenced by the alternative selected.
- Occurrence of a specific future is not known with certainty, even though certainty is often assumed for analysis purposes.

Decision Analysis Example 1

- Newsvendor Phyllis sells newspapers at the corner of Kirkwood Avenue and Indiana Street, and each day she must determine how many newspapers to order.
- She pays $0.20 for each paper and sells the papers for $0.25 each. Newspapers that are unsold at the end of the day are worthless.
- Phyllis knows that each day she can sell between 6 and 10 papers, with each possibility being equally likely.
- Set the problem as a decision analysis problem.

Decision Analysis Example 1

- $S = \{6, 7, 8, 9, 10\}$ are the possible values for the states of nature (daily demand for papers)
- Probability of each of the demand values occurring is $P_6 = P_7 = P_8 = P_9 = P_{10} = 1/5$
- $A = \{6, 7, 8, 9, 10\}$ course of action (alternatives) to choose from (papers to order each day).
- If i papers are purchased and j papers are demanded, then i papers are purchased at a cost of \$0.20i, and min(i, j) papers are sold for \$0.25 each.
- Reward (in cents): $R_{ij} = 25i - 20i = 5i$ for $i \leq j$
- Reward (in cents): $R_{ij} = 25j - 20i$ for $i > j$

Decision Analysis Example 1

- Payoff (Reward) Table

Papers Ordered (A_i)	Papers Demanded (S_j)				
	6	7	8	9	10
6	30	30	30	30	30
7	10	35	35	35	35
8	-10	15	40	40	40
9	-30	-5	20	45	45
10	-50	-25	0	25	50

- Why we did not consider that Phyllis would order 1, 2, 3, 4, 5, or more than 10 papers?
  - Answer involves the idea of dominated alternative(s).

Dominated Alternatives

- An alternative A_{i1} is dominated by an alternative A_{i2} if for all S_j included in S, the reward $R_{i1j} \leq R_{i2j}$, and for some state S_j, $R_{i1j} < R_{i2j}$.
  - If action A_{i1} is dominated, than in no state of the world is A_{i1} better than A_{i2}, and in at least one state of the world A_{i1} is inferior to A_{i2}.
- Generally, if an action A_i is dominated, there is no reason to choose it.
- Also called dominance criterion or elimination check.

Decision Analysis Example 1

- Enlarged Payoff (Reward) Table

Papers Ordered (A_i)	**Papers Demanded (S_j)**				
	6	7	8	9	10
1	5	5	5	5	5
2	10	10	10	10	10
3	15	15	15	15	15
4	20	20	20	20	20
5	25	25	25	25	25
6	**30**	**30**	**30**	**30**	**30**
7	**10**	**35**	**35**	**35**	**35**
8	**-10**	**15**	**40**	**40**	**40**
9	**-30**	**-5**	**20**	**45**	**45**
10	**-50**	**-25**	**0**	**25**	**50**
11	-70	-45	-20	5	30

Decision Analysis Example 2

- A computer systems firm has the opportunity to bid on two related contracts being advertised by a municipality.
  - The first pertains to the selection and installation of hardware for a central computing facility together with required software.
  - The second involves the development of a distributed computing network involving the selection and installation of hardware and software.
  - The firm may be awarded either contract C_1 or contract C_2, or both C_1 and C_2. Thus, there are three possible futures.

Decision Analysis Example 2

- Careful consideration of the possible approaches leads to the identification of five alternatives.
  - The first is for the firm to subcontract the hardware selection and installation, but to develop the software itself.
  - The second is for the firm to subcontract the software development, but to select and install the hardware itself.
  - The third is for the firm to handle both the hardware and software tasks itself.
  - The fourth is for the firm to bid jointly with a partner firm on both the hardware and software projects.
  - The fifth alternative is for the firm to serve only as project manager, subcontracting all hardware and software tasks.

Decision Analysis Example 2

- With the possible futures and various alternatives identified, the next step is to determine payoff values.
  - Also to be determined are the probabilities for each of the three futures, where the sum of these probabilities must be unity.
  - Suppose that these determinations lead to the profits and probabilities given in the table.

Decision Analysis Example 2

- Dominance Criterion.

Alternatives	Future Probability		
	C_1 (0.3)	C_2 (0.2)	$C_1 + C_2$ (0.5)
A_1	100	100	400
A_2	-200	150	600
A_3	0	200	500
A_4	100	300	200
A_5	-400	100	200

A_5 is dominated by all other four alternatives, so it should be eliminated.

Decisions Analysis Under Certainty

- Assumption that there is only one future for each alternative.
  - It is not claimed that knowledge about the future is known.
  - Suppression of risk and uncertainty is done for the purpose of simplifying reality to gain insight.
  - Insight can assist greatly in decision making, provided that its shortcomings are recognized and accommodated.

- A single future, which is assumed to occur with certainty, carries a probability of unity.
- All other futures are suppressed and carry probabilities of zero.

Decisions Analysis Under Certainty

- If alternatives are equal in all other respects, the decision maker chooses the alternative that minimizes cost or maximizes profit.
  - Minimize cost: min$\{R_i\}$ for i = 1, 2, …, m
  - Maximize profit: max$\{R_i\}$ for i = 1, 2, …, m

Alternative A_i	State of Nature (Future)
	S_j
A_1	R_1
A_2	R_2
…	…
A_m	R_m

Decision Analysis Under Risk

- There is little assurance that anticipated or predicted futures will coincide with actual futures.
- Decision making under risk occurs when the decision maker does not suppress acknowledged ignorance about the future but makes it explicit through the assignment of probabilities.

Decision Analysis Under Risk

- Aspiration Level Criterion.
  - Some form of aspiration level exists in most personal and professional decision making.
  - An aspiration level is some desired level of achievement such as profit, or some undesirable result level to be avoided, such as loss.

- Aspiration level criterion involves selecting some level of achievement that is to be met, followed by a selection of that alternative which maximizes the probability of achieving the stated aspiration level.

Decision Analysis Example 1

- Aspiration Level Criterion.
  - Set a minimum aspiration level for profit: 30
  - Set a maximum aspiration level for loss: 10

Papers Ordered (A_i)	**Papers Demanded (S_j)**				
	6	7	8	9	10
6	30	30	30	30	30
7	10	35	35	35	35
8	-10	15	40	40	40
9	-30	-5	20	45	45
10	-50	-25	0	25	50

A_6, A_7 and A_8 qualify

Decision Analysis Example 2

- Aspiration Level Criterion.
  - Set a minimum aspiration level for profit: 400
  - Set a maximum aspiration level for loss: 100

Alternatives	**Future / Probability**		
	C_1 (0.3)	C_2 (0.2)	$C_1 + C_2$ (0.5)
A_1	100	100	400
A_2	-200	150	600
A_3	0	200	500
A_4	100	300	200

A_1 and A_3 qualify

Decision Analysis Under Risk

- Most Probable Future Criterion.
  - Basic human tendency is to focus on the most probable outcome from among several that could occur.
  - All except for the most probable future are disregarded.

- Works well when the most probable future has a significantly high probability so as to partially dominate.

Decision Analysis Example 1

- Most Probable Future Criterion.
  - All futures have the same probability of occurrence.

Papers Ordered (A_i)	Papers Demanded (S_j)				
	6	7	8	9	10
6	30	30	30	30	30
7	10	35	35	35	35
8	-10	15	40	40	40
9	-30	-5	20	45	45
10	-50	-25	0	25	50

Decision Analysis Example 2

- Most Probable Future Criterion.
  - All, except the most probable future, are disregarded.
  - Choose the alternative that optimizes the outcome.

Alternatives	Future / Probability		
	C_1 (0.3)	C_2 (0.2)	$C_1 + C_2$ (0.5)
A_1	100	100	400
A_2	-200	150	600
A_3	0	200	500
A_4	100	300	200

A_2 is selected.

Can also be used to select between A_1 and A_3. Choose A_3.

Decision Analysis Under Risk

- Expected Value Criterion.
  - Decision makers strive to make choices that will maximize expected profit or minimize expected loss.
  - Viewed with caution when the payoff consequences of possible outcomes are disproportionately large, making a result that deviates from the expected outcome a distinct possibility.

- Requires weighing all payoffs by their probabilities of occurrence.
  - These weighed payoffs are them summed across all futures for each alternative.

Decision Analysis Example 1

- Expected Value Criterion.
  - Maximize expected profit or minimize expected loss.

Papers Ordered (A_i)	Papers Demanded (S_j)				
	6	7	8	9	10
6	30	30	30	30	30
7	10	35	35	35	35
8	-10	15	40	40	40
9	-30	-5	20	45	45
10	-50	-25	0	25	50

A_6: (30)(0.2)+(30)(0.2)+(30)(0.2)+(30)(0.2)+(30)(0.2)=**30**

A_7: (10)(0.2)+(35)(0.2)+(35)(0.2)+(35)(0.2)+(35)(0.2)=**30**

A_8: (-10)(0.2)+(15)(0.2)+(40)(0.2)+(40)(0.2)+(40)(0.2)=25

A_9: (-30)(0.2)+(-5)(0.2)+(20)(0.2)+(45)(0.2)+(45)(0.2)=15

A_{10}: (-50)(0.2)+(-25)(0.2)+(0)(0.2)+(25)(0.2)+(50)(0.2)=0

Decision Analysis Example 2

- Expected Value Criterion.
  - Maximize expected profit or minimize expected loss.

Alternatives	Future / Probability		
	C_1 (0.3)	C_2 (0.2)	$C_1 + C_2$ (0.5)
A_1	100	100	400
A_2	-200	150	600
A_3	0	200	500
A_4	100	300	200

- A_1: 100(0.3) + 100(0.2) + 400(0.5) = 250
- A_2: -200(0.3) + 150(0.2) + 600(0.5) = 270
- A_3: 0(0.3) + 200(0.2) + 500(0.5) = 290 - Selected
- A_4: 100(0.3) + 300(0.2) + 200(0.5) = 190

Decision Analysis Example 1

- Comparison of Decisions.
  - There is no one best selection when these criteria are used for decision making under risk.
  - Decision depends on the decision criterion selected.

  - Aspiration level criterion: A_6, A_7, or A_8
  - Most probable future criterion: not useful
  - Expected value criterion: A_6 or A_7
  - Overall, best alternative: Cannot decide (maybe A_6 or A_7)

Decision Analysis Example 2

- Comparison of Decisions.
  - There is no one best selection when these criteria are used for decision making under risk.
  - Decision depends on the decision criterion selected.

  - Aspiration level criterion: A_1 or A_3
  - Most probable future criterion: A_2
  - Expected value criterion: A_3
  - Overall, best alternative: A_3

Decision Analysis Under Uncertainty

- When probabilities are not available for assignment to future events.
  - Inappropriate or impossible to assign probabilities to the several futures identified for a given decision situation.
  - No meaningful data are available from which probabilities may be developed.
  - Decision maker may be unwilling to assign a subjective probability, as is often when the future could prove to be unpleasant.

Decision Analysis Under Uncertainty

- Laplace Criterion.
  - Each possible state of nature is as likely to occur as any other.
  - No stated bias for one state of nature to be more likely than any other.
  - Called Laplace principle or the principle of insufficient reason based on the philosophy that nature is assumed to be indifferent.

- Probability of the occurrence of each future state of nature is assumed to be $1/n$, where n is the number of possible future states.

Decision Analysis Example 1

- Laplace Criterion.
  - The actual assumption of newsvendor demand problem.

Alternative	Expected Reward
A_6	(30 + 30 + 30 + 30 + 30)/5 = 30
A_7	(10 + 35 + 35 + 35 + 35)/5 = 30
A_8	(-10 + 15 + 40 + 40 + 40)/5 = 25
A_9	(-30 – 5 + 20 + 45 + 45)/5 = 15
A_{10}	(-50 – 25 + 0 + 25 + 50)/5 = 0

Alternatives A_6 or A_7 are selected.

Decision Analysis Example 2

- Laplace Criterion.
  - Firm is unwilling to differentiate between the likelihood of acquiring contract C_1 , contract C_2, or contract C_1 and C_2.

Alternative	Average profit
A_1	(100 + 100 + 400)/3 = 200
A_2	(-200 + 150 + 600)/3 = 183
A_3	(0 + 200 + 500)/3 = 233
A_4	(100 + 300 + 200)/3 = 200

Alternative A_3 is selected.

Decision Analysis Under Uncertainty

- Maximin Criteria.
  - Based on an extremely pessimistic view of the outcome.
  - Justified if it is judged that nature will do its worst.
  - Leads to the alternative that assures the best of the worst possible outcomes.

- For each action, determine the worst outcome (smallest reward).
- Choose action A_i with the largest value of $\min\{R_{ij}\}$
  - $\max(\min R_{ij})$

Decision Analysis Example 1

- Maximin Criteria.

Papers Ordered (A_i)	Papers Demanded (S_j)					Maximin Criterion
	6	7	8	9	10	
6	30	30	30	30	30	30
7	10	35	35	35	35	10
8	-10	15	40	40	40	-10
9	-30	-5	20	45	45	-30
10	-50	-25	0	25	50	-50

Alternative A_6 is selected.

Decision Analysis Example 2

- Maximin Criteria.

Alternatives	Future / Probability			Maximin Criterion
	C_1 (0.3)	C_2 (0.2)	$C_1 + C_2$ (0.5)	
A_1	100	100	400	100
A_2	-200	150	600	-200
A_3	0	200	500	0
A_4	100	300	200	100

Alternatives A_1 or A_4 are selected.

Decision Analysis Under Uncertainty

- Maximax Criteria.
  - Based on an extremely optimistic view of the outcome.
  - Justified if it is judged that nature will do its best.
  - Leads to the alternative that assures the best of the best possible outcomes.
- For each action, determine the best outcome (largest reward).
- Choose action A_i with the largest value of $\max\{R_{ij}\}$
  - $\max(\max R_{ij})$

Decision Analysis Example 1

- Maximax Criteria.

Papers Ordered (A_i)	Papers Demanded (S_j)					Maximax Criterion
	6	7	8	9	10	
6	30	30	30	30	30	30
7	10	35	35	35	35	35
8	-10	15	40	40	40	40
9	-30	-5	20	45	45	45
10	-50	-25	0	25	50	50

Alternative A_{10} is selected.

Decision Analysis Example 2

- Maximax Criteria.

Alternatives	Future / Probability			Maximax Criterion
	C_1 (0.3)	C_2 (0.2)	$C_1 + C_2$ (0.5)	
A_1	100	100	400	400
A_2	-200	150	600	600
A_3	0	200	500	500
A_4	100	300	200	300

Alternative A_2 is selected.

Decision Analysis Under Uncertainty

- Minimax Regret Criterion.
  - Uses the concept of opportunity to arrive at a decision.
  - For each possible state of the world S_j, find an action $i^*(j)$ that maximizes R_{ij}.
  - $i^*(j)$ is the best possible action to choose if the state of the world is actually S_j.
  - For any action A_i and state S_j, the opportunity loss or regret for A_i in S_j is $R_{i^*(j)\,j} - R_{ij}$.

Decision Analysis Example 1

- Minimax Regret Criterion.
  - If j = 7 papers are demanded, the best decision is to order $i^*(7)$ = 7 papers, yielding a profit of $r_{77} = 7(25) - 7(20) = 35$.
  - Suppose we choose to order i = 6 papers.

    Since $r_{67} = 6(25) - 6(20) = 30$, the opportunity loss or regret for i = 6 and j = 7 is 35 – 30 = 5.
  - Thus, if we order 6 papers and 7 are demanded, we realize that by making the optimal choice (ordering 7) for the actual state of the world (7 papers demanded), we would have done 5 better as reward than we did by ordering 6.

Decision Analysis Example 1

- Minimax Regret Criterion.

Papers Ordered (A_i)	Papers Demanded (S_j)					Minimax Regret Criterion
	6	7	8	9	10	
6	30-30=0	35-30=5	40-30=10	45-30=15	50-30=20	20
7	30-10=20	35-35=0	40-35=5	45-35=10	50-35=15	20
8	30+10=40	35-15=20	40-40=0	45-40=5	50-40=10	40
9	30+30=60	35+5=40	40-20=20	45-45=0	50-45=5	60
10	30+50=80	35+25=60	40-0=40	45-25=20	50-50=0	80

Alternative A_6 and A_7 are selected.

Decision Analysis Example 2

- Minimax Regret Criterion.

Alternatives	Future / Probability			Minimax Regret Criterion
	C_1 (0.3)	C_2 (0.2)	$C_1 + C_2$ (0.5)	
A_1	100-100=0	300-100=200	600-400=200	200
A_2	100+200=300	300-150=150	600-600=0	300
A_3	100-0=100	300-200=100	600-500=100	100
A_4	100-100=0	300-300=0	600-200=400	400

Alternative A_3 is selected.

Decision Analysis Under Uncertainty

- Hurwicz Criterion.
  - Most decision makers have a degree of optimism or pessimism somewhere between the extremes.
  - Considers relative optimism and pessimism.
  - Select index of optimism, α, $0 \leq \alpha \leq 1$
  - $\alpha = 0$ indicates pessimism
  - $\alpha = 1$ indicates optimism
  - R_{ij} = payoff for the ith alternative and the jth state

$$\max_i \left\{ \alpha \left[\max_j R_{ij} \right] + (1-\alpha) \left[\min_j R_{ij} \right] \right\}$$

Decision Analysis Under Uncertainty

- Hurwicz Criterion discussion.
  - When α = 0, Hurwicz rule becomes maximin rule.

$$\max_i \left\{ 0 \left[\max_j R_{ij} \right] + (1-0) \left[\min_j R_{ij} \right] \right\} = \max_i \left[\min_j R_{ij} \right]$$

  - When α = 1, Hurwicz rule becomes maximax rule.

$$\max_i \left\{ 1 \left[\max_j R_{ij} \right] + (1-1) \left[\min_j R_{ij} \right] \right\} = \max_i \left[\max_j R_{ij} \right]$$

Decision Analysis Example 1

- Hurwicz Criterion.
  - Select index of optimism, α = 0.2

Alternative	Expected reward
A_6	(0.2)(30) + (0.8)(30) = 30
A_7	(0.2)(35) + (0.8)(10) = 15
A_8	(0.2)(40) + (0.8)(-10) = 0
A_9	(0.2)(45) + (0.8)(-30) = -15
A_{10}	(0.2)(50) + (0.8)(-50) = -30

Select A_6

Decision Analysis Example 2

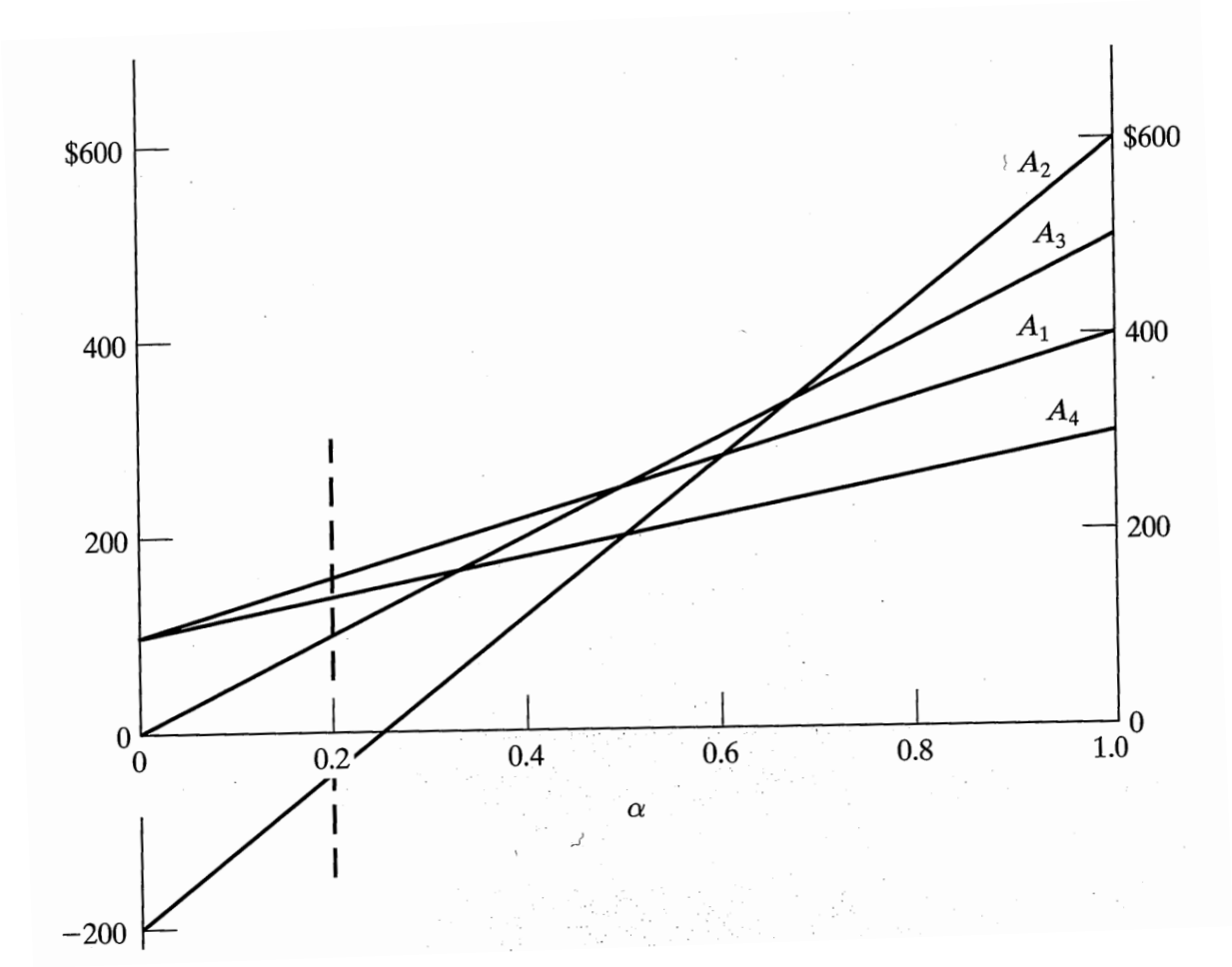
- Hurwicz Criterion.
  - Select index of optimism, α = 0.2

Alternative	Expected reward
A_1	(0.2)(400) + (0.8)(100) = 160
A_2	(0.2)(600) + (0.8)(-200) = -40
A_3	(0.2)(500) + (0.8)(0) = 100
A_4	(0.2)(300) + (0.8)(100) = 140

Select A_1

Decision Analysis Example 2

- Hurwicz Criterion.



Decision Analysis Example 2

- Comparison of Decisions.
  - Laplace criterion: A_6 or A_7
  - Maximin criterion: A_6
  - Maximax criterion: A_{10}
  - Minimax regret: A_6 or A_7
  - Hurwicz criterion: A_6
  - Choice of rule is based on attitude toward uncertainty (pessimistic or optimistic) and utility function.

Decision Analysis Example 2

- Comparison of Decisions.
  - Laplace criterion: A_3
  - Maximin criterion: A_1 or A_4
  - Maximax criterion: A_2
  - Hurwicz criterion: A_1
  - Minimax regret: A_4
  - Choice of rule is based on attitude toward uncertainty (pessimistic or optimistic) and utility function.

More Decision Models…

Decision Analysis Under Uncertainty

- Maximin Criterion.
  - Determine the minimum profit associated with each alternative and then select the alternative that maximizes the minimum profit.

- For each action, determine the worst outcome.
- Choose action A_i with the largest value of $\min(R_{ij})$
  - $\max[\min(R_{ij})]$

Decision Analysis Example

- Maximin Criterion.
  - Assume R_{ij} are profits.

Alternatives	Future				Maximin Criterion
	S_1	S_2	S_3	S_4	
A_1	6	1	1	5	1
A_2	4	0	2	3	0
A_3	5	4	5	2	2

Alternatives A_3 is selected.

Decision Analysis Under Uncertainty

- Minimax Criterion.
  - Determine the maximum cost associated with each alternative and then select the alternative that minimizes the maximum cost.

- For each action, determine the worst outcome.
- Choose action A_i with the smallest value of $\max(R_{ij})$
  - $\min[\max(R_{ij})]$

Decision Analysis Example

- Minimax Criterion.
  - Assume R_{ij} are costs.

Alternatives	Future				Minimax Criterion
	S_1	S_2	S_3	S_4	
A_1	6	1	1	5	6
A_2	4	0	2	3	4
A_3	5	4	5	2	5

Alternatives A_2 is selected.

Decision Analysis Under Uncertainty

- Maximax Criterion.
  - Determine the maximum profit associated with each alternative and then select the alternative that maximizes the maximum profit.

- For each action, determine the best outcome.
- Choose action A_i with the largest value of $\max(R_{ij})$
  - $\max[\max(R_{ij})]$

Decision Analysis Example

- Maximax Criterion.
  - Assume R_{ij} are profits.

Alternatives	Future				Maximax Criterion
	S_1	S_2	S_3	S_4	
A_1	6	1	1	5	6
A_2	4	0	2	3	4
A_3	5	4	5	2	5

Alternatives A_1 is selected.

Decision Analysis Under Uncertainty

- Minimin Criterion.
  - Determine the minimum cost associated with each alternative and then select the alternative that minimizes the minimum cost.

- For each action, determine the best outcome.
- Choose action A_i with the smallest value of $\min(R_{ij})$
  - $\min[\min(R_{ij})]$

Decision Analysis Example

- Minimin Criterion.
  - Assume R_{ij} are costs.

Alternatives	Future				Minimin Criterion
	S_1	S_2	S_3	S_4	
A_1	6	1	1	5	1
A_2	4	0	2	3	0
A_3	5	4	5	2	2

Alternatives A_2 is selected.